



Fig. 11: Tactical high-energy laser

TRW, high-energy laser weapon system from TRW, Boeing's airborne laser and advanced tactical laser, tactical high-energy laser, mobile tactical high-energy laser, Raytheon's Laser Phalanx and laser weapon system.

High-energy laser, jointly developed by two German companies Diehl and LFK, is a short-range tracked vehicle-mounted system for use as an air-defence system against low-flying, high-performance aircraft, missiles and attack helicopters. The system is configured around a gas dynamic CO₂ laser emitting at 10.6 microns. It has associated target-acquisition and tracking sensors.

Another air-defence laser system is TRW's general-area defence integrated anti-missile laser system. This system, too, is a short-range complement to surface-to-air missile defence designed to engage discrete ballistic threats at longer ranges. The system generates a 400kW laser beam, which is delivered through a 0.7-metre beam pointer/tracker.

One of the very early directed energy laser systems, a technology demonstrator, is TRW's mid infrared chemical laser. It is a 2.2MW

CW deuterium-fluoride laser with a maximum lasing duration of 70 seconds. It uses a 1.8-metre Sealite pointing and tracking device. The system was reported to carry out trials against different types of targets at White Sands Missile Range in New Mexico. In one of the tests conducted in 1996, a small fraction of laser power was used to destroy a 122mm short-range artillery rocket in flight. The laser beam was locked to the target for 15 seconds. It is also reported to have been tested against sea-skimming missiles.

Yet another recent system from TRW is high-energy laser weapon system, which is again a chemical laser using ethylene, hydrogen and fluorinated nitrogen as the active medium. The system is based on their earlier experience gained from building mid-infrared chemical laser. The system has an integral electro-optic tracker.

Another laser based directed energy weapon system in serious contention till recent past, particularly for operation from an aerial platform, is the famous airborne laser. It uses a 1.2-megawatt chemical oxy-iodine laser emitting at 1.315 microns, generated by six

chemical oxy-iodine laser modules of 200 kilowatts each. It is configured on Boeing 747-400 freighter aircraft (Fig. 10).

Airborne laser is capable of destroying a ballistic missile in boost phase. In operation, the aircraft patrols friendly air space. After an enemy missile launch is detected by a variety of sensors, this information is relayed to the aircraft configured as high-power laser system for further action.

Advance tactical laser uses 80kW chemical oxy-iodine laser and is mounted on a modified Boeing C-130H Hercules aircraft, with the most obvious visual difference being a rotating turret protruding from the aircraft's underside through a hole. This chemical laser is similar to the one developed for Airborne Laser programme with much lower output power.

Advance tactical laser is envisioned to offer mobility of a small aircraft, high-resolution imagery for target identification and the ability to localise damage to a small area of less than 30.5cm (1-foot) in diameter from a range of eight to ten kilometres.

Another well-known laser based DEW system is Northrop Grumman's tactical high-energy laser. The laser is built in two configurations: baseline static high-energy laser and relocateable mobile version mobile tactical high-energy laser.

Tactical high-energy laser systems (Fig. 11) are point defence weapon systems designed to engage and destroy artillery rockets, artillery shells, mortar rounds and low-flying aircraft. The system uses a deuterium fluoride laser operating at 3.8µm. Tactical high-energy laser demonstrator was successfully tested repeatedly between 2000 and 2004, destroying a number of 122mm and 160mm Katyusha rockets, multiple artillery shells and mortar rounds, including a salvo attack by mortar.